Planning Controls

Planning Week

1

API: healthy

SupplyNet AI

Supply network planning, optimization, scenario simulation, and agentic decision support.

Network Overview

Suppliers	Distribution Centers	Customer Regions	Inbound Lanes
25	4	8	100

Active Suppliers	Available Inbound Lanes	Outbound Lanes	Available Outbound Lanes
23	96	32	32

Week 1 Plan Comparison

Naive Cost	Optimized Cost	Cost Savings	Solver Status
$1,121,559	$1,033,747	$87,812 ↑ 7.83%	OPTIMAL

Service Performance

Naive Fulfillment	Optimized Fulfillment	Fulfillment Change
100.00%	100.00%	0.00%

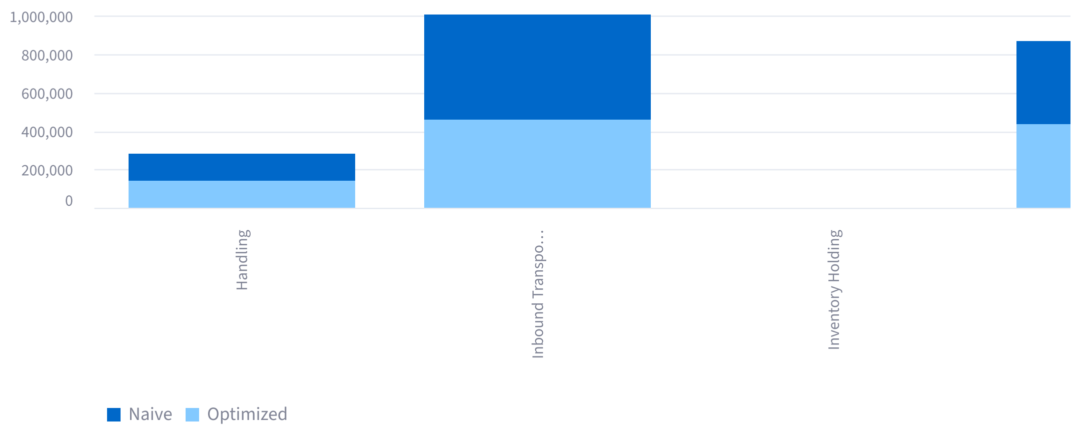
Naive Unmet Demand	Optimized Unmet Demand	Total Demand
0	0	9,525

Container Utilization

Naive Utilization	Optimized Utilization	Utilization Change
97.36%	99.88%	2.53%

Cost Breakdown

	Cost Component	Naive
0	Inbound Transportation	$549,545
1	Outbound Transportation	$431,233
2	Handling	$140,780
3	Inventory Holding	$0
4	Shortage	$0



Optimized Network Plan

Load Optimized Allocation Details

AI Scenario Analyst

Describe an operational disruption in natural language. SupplyNet AI converts it into a validated scenario, reruns the deterministic optimization model, and evaluates the financial and service impact.

Example scenarios

Custom scenario

Describe scenario

Increase West demand by 20% in week 1

Analyze Scenario

Scenario analyzed successfully.

Interpreted Scenario

{
"scenario_type" : "demand_surge"
"week" : 1
"supplier_id" : NULL

```
  "dc_id" : NULL
  "region_id" : "REG_WEST"
  "percentage" : 0.2
}
```


Financial Impact

Baseline Cost	Scenario Cost	Cost Impact
$1,033,747	$1,078,549	$44,802 ↑ 4.33%

Service Impact

Baseline Fulfillment	Scenario Fulfillment	Scenario Unmet Demand
100.00%	100.00%	0

Recommendation

The network absorbs the disruption while maintaining 100.00% fulfillment. Cost increases by only 4.33%, so the optimized rerouting is operationally viable.

How SupplyNet AI analyzed this scenario

Decision workflow

1. Natural-language scenario received
2. Scenario parsed into structured parameters
3. Parameters validated with Pydantic
4. Base network copied into a scenario network
5. Requested disruption applied
6. OR-Tools MILP optimizer rerun
7. Scenario compared with baseline optimization
8. Business recommendation generated

The agent does not calculate shipment quantities itself. Shipment and routing decisions are produced by the deterministic optimization engine.

Planning Insight

The optimized plan reduces total network cost by 7.83% relative to the rule-based planning baseline.

Optimization maintains or improves customer fulfillment while minimizing network cost.